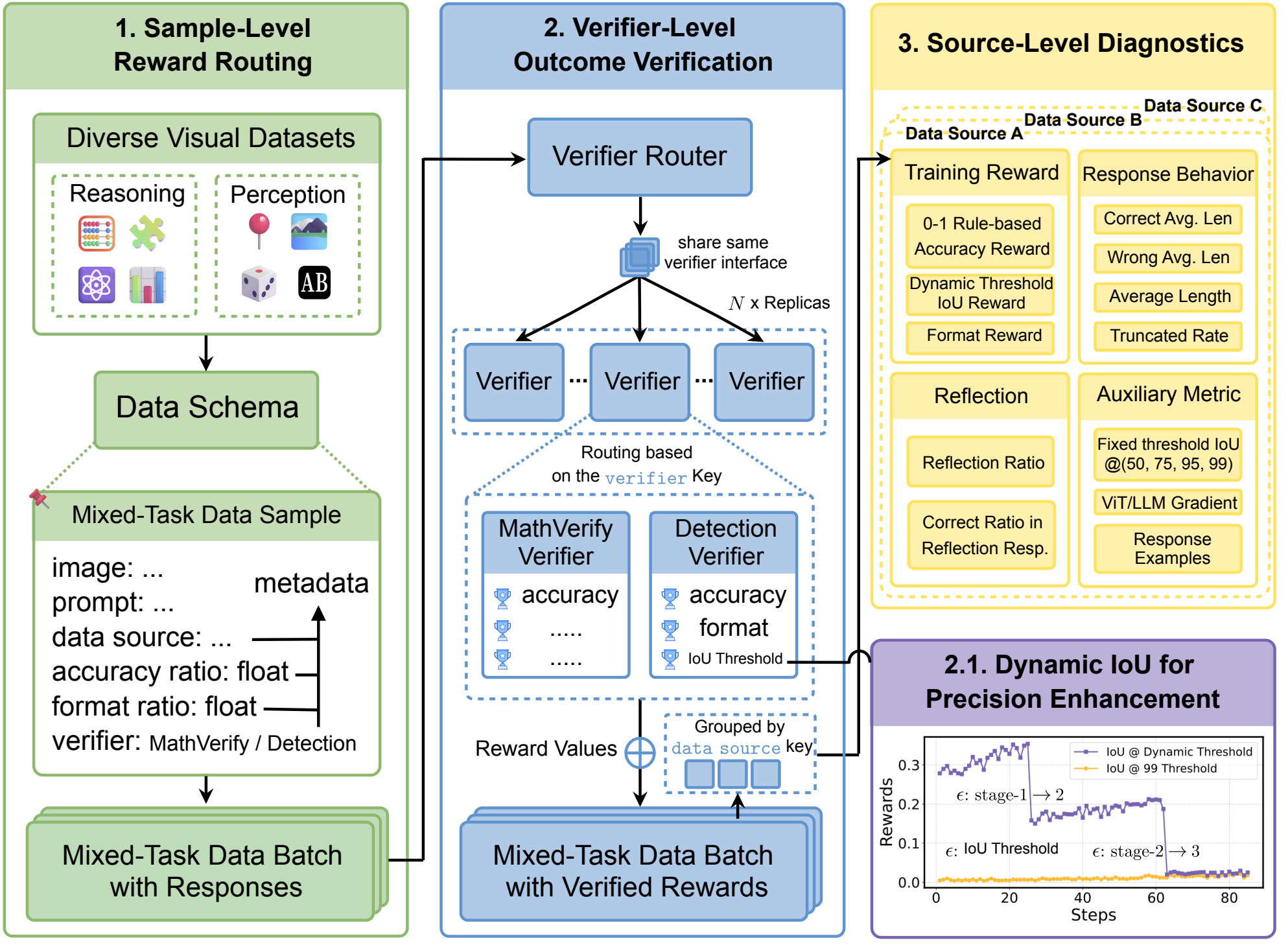
1. Sample-Level Reward Routing
Diverse Visual Datasets
Reasoning
Perception
AB
Data Schema
Mixed-Task Data Sample
image: ...
prompt: ...
data source: ...
accuracy ratio: float
format ratio: float
verifier: MathVerify / Detection
metadata
Mixed-Task Data Batch with Responses
2. Verifier-Level Outcome Verification
Verifier Router
share same verifier interface
N x Replicas
Verifier
Verifier
Verifier
Routing based on the verifier Key
MathVerify Verifier
accuracy
.....
.....
Detection Verifier
accuracy
format
IoU Threshold
Reward Values
Grouped by data source key
Mixed-Task Data Batch with Verified Rewards
3. Source-Level Diagnostics
Data Source C
Data Source B
Data Source A
Training Reward
0-1 Rule-based Accuracy Reward
Dynamic Threshold IoU Reward
Format Reward
Response Behavior
Correct Avg. Len
Wrong Avg. Len
Average Length
Truncated Rate
Reflection
Reflection Ratio
Correct Ratio in Reflection Resp.
Auxiliary Metric
Fixed threshold IoU @(50, 75, 95, 99)
ViT/LLM Gradient
Response Examples
2.1. Dynamic IoU for Precision Enhancement
IoU @ Dynamic Threshold
IoU @ 99 Threshold
ϵ: stage-1 → 2
ϵ: IoU Threshold
ϵ: stage-2 → 3
Rewards
0.0
0.1
0.2
0.3
0
20
40
60
80
Steps